



Huangqi Jiuni decoction prevents acute kidney injury induced by severe burns by inhibiting activation of the TNF/NF- κ B pathway

Wanxuan Hu^{a,1}, Jie Zhao^{b,c,1}, Yuxin Hu^c, Shuai Song^d, Xulin Chen^a, Yexiang Sun^{a,*}

^a Department of Burn, The First Affiliated Hospital of Anhui Medical University, Hefei, Anhui, 230022, PR China

^b Department of Chinese Medicine, The First Affiliated Hospital of Anhui Medical University, No.218, Jixi Road, Shushan District, Hefei, Anhui, 230032, PR China

^c Department of Chinese Integrative Medicine, Anhui Medical University, No. 80, Meishan Road, Shushan District, Hefei, Anhui, 230032, PR China

^d Department of Pharmacy, The First Affiliated Hospital of Anhui Medical University, Hefei, 230022, PR China

ARTICLE INFO

Handling Editor: Dr. Thomas Efferth

Keywords:

HQJND

LC-MS/MS

Network pharmacology

TNF/NF- κ B pathway

Severe burns

Acute kidney injury

ABSTRACT

Ethnopharmacological relevance: Huangqi Jiuni decoction (HQJND) is a prescription for the treatment of severe burns provided based on traditional Chinese and Western medicine, which is created by the First Affiliated Hospital of Anhui Medical University. It consists of 12 herbs and has been used clinically for decades. It has greatly shortened the course of the disease, but the mechanism by which HQJND treats the disease still remains unclear.

Aim of the study: Hence, the objective of this investigation was to utilize modern pharmacological tools to demonstrate the efficacy and mechanism of HQJND in the treatment of acute kidney injury (AKI) caused by severe burns.

Materials and methods: In this study, the chemical constituents in HQJND were first examined using liquid chromatography tandem mass spectrometry (LC-MS/MS). Then, by using network pharmacology, we screened the targets of drug and disease action, and predicted the signaling pathways acting in the course of drug treatment of disease. Finally, we attempted to verify the efficacy of the drug and explored its therapeutic mechanism after the establishment of an animal model, herbal gavage treatment, collection of rat kidneys and serum for renal function, quantitative real-time Polymerase Chain Reaction (RT-qPCR), Western Blotting (WB), Hematoxylin and eosin (HE) staining and Immunohistochemistry (IHC).

Results: The 14 important active ingredients in HQJND was analyzed by liquid chromatography tandem mass spectrometry, while network pharmacology screening was performed to identify 353 disease-associated marker genes and 286 drug targets, finally identifying the TNF/NF- κ B (tumor necrosis factor/nuclear factor kappa-B) signaling site: the key pathway of burn-induced acute kidney injury when HQJND intervened. The serum renal function and histopathology of rats demonstrated that the use of HQJND significantly improved the renal function in severe burns. RT-qPCR and WB confirmed that the TNF/NF- κ B signaling pathway was activated in the Model group of rats, and HQJND could curb the signaling pathway because it moderated the expressions of key proteins in the process.

Conclusion: Based on modern pharmacology, we explored an effective herbal preparation to ameliorate the impairment of renal function after severe burns, which is most likely to function through the TNF/NF- κ B signaling pathway.

1. Introduction

So far, burns have remained as an unavoidable surgical injury (Mayhall, 2003). A significant proportion of burned patients suffer from deep and large-sized tissue damage which is likely to be accompanied

with various organ dysfunctions and even cause deaths (Li et al., 2017). Burned patients with AKI, the ubiquitous complication in patients with severe burns, have an increased risk not only for death, but also chronic kidney disease (Folkestad et al., 2020). Currently, the primary treatment for burn-induced AKI is the prevention of shock through rehydration while renal replacement therapy is used for more critical patients

* Corresponding author.

E-mail address: sunyexiang@163.com (Y. Sun).

¹ These authors contributed equally to this research.

Abbreviations:

HQJND	Huangqi Jiuni decoction
LC-MS/MS	liquid chromatography tandem mass spectrometry
RT-qPCR	quantitative real-time Polymerase Chain Reaction
WB	Western Blotting
AKI	acute kidney injury
H&E	Hematoxylin and eosin
IHC	immunohistochemistry
NF-κB	Nuclear factor kappa B
TNF-α	tumor necrosis factor-α
TCM	Traditional Chinese Medicine
KEGG	kyoto encyclopedia of genes and genomes
GO	Gene ontology
LPS	lipopolysaccharide
KIM1	Kidney injury molecule 1
DAB	3,3-N-Diaminobenzidine Tetrahydrochloride
SEM	standard error of mean
PPI	protein-protein interaction
TBST	Tris-Buffered Saline Tween-20
ANOVA	Analysis of Variance
IL6	Interleukin-6
IL1B	Interleukin-1 beta
IL2	Interleukin-2

CXCL8	C-X-C motif chemokine ligand 8
ICAM1	Intercellular adhesion molecule 1
IL10	Interleukin-10
MMP2	Matrix metalloproteinase-2
EGF	Pro-epidermal growth factor
CXCL2	C-X-C motif chemokine 2
IFNG	Interferon gamma
IL4	Interleukin-4
VCAM1	Vascular cell adhesion molecule-1
BCL2	B-cell lymphoma 2
CASP3	Caspase-3
CAV1	Caveolin-1
COL3A1	Collagen alpha-1(III) chain
IRF1	Interferon regulatory factor 1
BAX	BCL2 associated X
HMOX1	Heme oxygenase 1
STAT3	Signal transducer and activator of transcription 3
FOS	Fos proto-oncogene
TNFR1	Tumor necrosis factor receptor superfamily member 1A
PP65	Phospho-Nuclear factor-kB p65
P65	Nuclear factor-kB p65
PIKBα	phospho-NF-κappa-B inhibitor alpha
IKBα	NF-κappa-B inhibitor alpha
GAPDH	Glyceraldehyde-3-phosphate dehydrogenase

Table 1
Component herbs of HQJND.

Chinese name	Scientific name	Amount (g)
Huang Qi	<i>Astragalus mongholicus</i> Bunge	50.0
Dang Shen	<i>Codonopsis pilosula</i> (Franch.) Nannf.	25.0
Bai Zhu	<i>Atractylodes macrocephala</i> Koidz.	15.0
Dang Gui	<i>Angelica sinensis</i> (Oliv.) Diels	25.0
Sheng Di Huang	<i>Rehmannia glutinosa</i> (Gaertn.) DC.	25.0
Bai Shao	<i>Paeonia lactiflora</i> Pall.	25.0
Dan Shen	<i>Salvia miltiorrhiza</i> Bunge	25.0
Wu Wei Zi	<i>Schisandra chinensis</i> Turcz. (Baill.)	15.0
Ze Xie	<i>Alisma plantago-aquatica</i> subsp. <i>orientale</i> (Sam.) Sam.	25.0
Che Qian Zi	<i>Plantago ovata</i> Forssk.	25.0
Zhu Ling	<i>Polyporus umbellatus</i> (Pers.) Fries	25.0
Fu Ling	<i>Poria cocos</i> (Schw.) Wolf	25.0

Table 2
Primer list for RT-qPCR analysis.

Gene name	Forward sequence (5'–3')	Reverse sequence (5'–3')
KIM1	CACTGTCTTCAGTCAACAC	TAAGAATCTACAGAGCCTGGAAG
GAPDH	CCGCATCTTCTGTGCAGTG	CGATACGGCCAAATCCGTTTC
P65	TTCAACATGGCAGACGACGA	AGGTATGGGCCATCTGTTGAC
TNFα	ATGGGCTCCCTCTCATCAGT	GCTTGGTGGTGTGTGCTACGAC
TNFR1	CCAAGTGCCACAAAGGAACC	CACACCTCGCAGACTGTTTC

(Lavrentieva et al., 2022). However, these methods have failed to eliminate clinical problems. That is because only rehydration is inadequate to restore hemodynamic stability since vascular permeability increases after the onset of severe burns. Another problem is that insufficient rehydration may lead to injury in renal ischemia (Niculae et al., 2022) but too much rehydration may trigger tissue edema and aggravate the injury (Prowle et al., 2014). In addition, there is no standardized guideline for the use of renal replacement therapy (Bagshaw et al., 2017), so improper application may aggravate hemodynamic instability (Villa et al., 2021). In this sense, traditional Chinese

medicine is proposed as a complementary therapy for AKI (Shaterzadeh-Yazdi et al., 2018), since many Chinese medicines are inherently renoprotective (Hui et al., 2021).

HQJND is a herbal combination that has long been used clinically for the treatment of severe burns combined with shock and sepsis. It has been recognized as National Invention Patent of China in 2023 (ZL, 2022 1 0926816.1) (Supplementary 1). Clinical observation has found that compared with the conventional burn treatment, it can reduce the amount of rehydration and wound exudation, and alleviate the inflammatory response. In a decade of clinical application, it has been found to significantly shorten the course of the disease, improve the clinical efficacy, and greatly improve the renal function of patients. Therefore, in a previous work, we investigated the beneficial influence of HQJND on renal function in scalded rabbits. We found that the mixture was able to reduce tissue damage by increasing the circulating blood volume of the kidneys, but the mechanism of how the drug specifically serve to treat kidney injury after burns is currently unknown. Thus, this study is proposed to further explore the efficacy and potential therapeutic mechanisms of HQJND on AKI. The HQJND consists of 12 herbs. Table 1 summarizes the chinese name, scientific name and weights of HQJND. The herbal names in the table have been validated by World Flora Online (www.worldfloraonline.org).

The components of Chinese medicines are characterized with complexity and each constituent acts on different target points, which makes it inconvenient to study the mechanism of Chinese medicines. In order to determine the mechanism by which a particular herbal combination treats a disease, researchers usually use methods such as chromatography-mass spectrometry, network pharmacology and molecular docking (Sharma and Yadav, 2022). LC-MS/MS is taken as an essential approach to characterizing the active ingredients of herbal medicines, aiming to control the safety and efficacy of herbal medicines from different sources through qualitative and quantitative analysis of active ingredients in drugs (Beccaria and Cabooter, 2020). Network pharmacology integrates multidisciplinary techniques such as bioinformatics, genetics and proteomics. It summarizes the therapeutic mechanism of a drug by collecting as much information as possible about the drug's target of action (Zhao et al., 2023). The number of drugs is immeasurable, so it would be resources-wasting to develop and

Table 3

Optimized MRM transitions and the corresponding parameter of 14 compounds.

Compound	Retention time (min)	Ionization mode	MRM transitions (Precursor-Product)	Q1 Pre bias (V)	Collision Energy	Q3 Pre bias (V)
Geniposidic acid	0.911	–	373.10 > 123.05	18	21	10
Paeoniflorin	0.932	–	479.05 > 121.05	11	23	18
Caffeic acid	0.948	–	179.15 > 135.05	12	25	21
Lobetyolin	1.031	–	329.05 > 187.05	24	27	26
Ferulic Acid	1.054	–	193.10 > 134.05	22	29	22
Azelaic Acid	1.133	–	187.15 > 168.90	19	19	10
Formononetin	1.66	–	267.05 > 223.05	13	32	21
Atractylenolide III	2.062	+	249.10 > 105.10	–12	–35	–25
Columbianadin	2.743	+	441.10 > 143.10	–12	–21	–21
Astragaloside IV	2.857	–	829.25 > 783.40	24	27	26
Tanshinone I	4.314	+	277.05 > 178.05	–10	–43	–19
TanshinonellA	6.023	+	295.00 > 206.10	–21	–37	–23
Schisandrin B	6.053	+	401.10 > 300.10	–25	–27	–21
Ergosterol	7.916	+	397.30 > 105.15	–25	–52	–22

study every drug. The advent of molecular docking effectively reduces this waste by allowing drug molecules and proteins to be simulated in a computer with different binding modes, allowing for faster discovery of drugs with potential applications (Crampon et al., 2022).

2. Materials and methods

2.1. LC-MS/MS analysis

Soak the herbs (all drugs were provided by the Chinese herbal pharmacy of the First Affiliated Hospital of Anhui Medical University) in water for 30 min, then pour them into a saucepan for decoction. Specifically, 800 ml of water was added to the decoction for the first time, then the decoction was brought to boil and then reduced heat, simmer for half an hour. Next, the decocted medicinal liquid was poured into a bowl to obtain 400 ml of medicinal liquid for the first time. Afterwards, 600 ml of water was poured into the saucepan for the second time to obtain 300 ml of liquid decoction. The liquid obtained twice was mixed to obtain 700 ml total volume of the drug. Finally, we concentrated the 700 ml of liquid into 100 ml of viscous concentrated drug by rotary evaporator and set aside for use.

Quantification of bioactive compounds in HQJND was performed on a Shimadzu LC-MS (8045) system (equipped with two LC-30A pumps, a SIL-30AC autosampler, a CTO-30A column oven and LCMS-8045 triple-quadrupole mass spectrometry). The chromatographic separation was achieved isocratically on Shim-pack GIST C18 column (100 mm × 2.0 mm i.d.; 2 μm) at 40 °C. Mobile phase consisted of methanol with formic acid (0.1% v/v) (A) and aqueous formic acid (0.1% v/v) (B), and the flow rate of 0.3 mL/min. The injection volume was 10 μl for quantitative analysis.

For the mass spectrometer, an electrospray ionization source (ESI) and multiple reaction monitoring (MRM) mode was equipped to acquire chromatography in positive and negative electrospray ionization with the final conditions as follows: nebulizer gas 3 L/h, drying gas 10 L/h, desolvation line (DL) temperature 300 °C, heat block temperature 300 °C, collision gas (argon) 230 kPa. The dwell time was set at 30 msec.

2.2. Network pharmacology

2.2.1. Screening of protein targets acted by HQJND active ingredients

In an attempt to identify the protein targets of the actual constituents of the HQJND, we referred to the database of Traditional Chinese Medicine Systems Pharmacology (TCMSP <https://old.tcmsp-e.com/tcmsp.php>) and collected active ingredients in HQJND. Furthermore, since Astragaloside IV, Rehmannioside D, Acteoside, Catalpol, Schisandrin B have many pharmacological effects, but the TCMSP database does not include the targets of these chemical components, so we added the targets of the above chemical components from PharmMapper Server (<http://www.lilab-ecust.cn/pharmmapper/>). The active ingredients in

herb are mainly screened by setting two restrictions: oral bioavailability (OB) ≥ 30% and drug-likeness (DL) ≥ 0.18. However, these selected standards are not totally fixed, for example, Astragaloside IV is not in this range, but we retained this active ingredient considering that it possesses important pharmacological effects and it can be really detected in the drug. Immediately after screening, we converted the obtained target information from protein names to canonical gene names with the help of Uniprot (<https://www.uniprot.org>).

2.2.2. Establishing drug-active ingredient-target networks

Cytoscape is a software that can analyze, edit and visualize data. Collected drug, active ingredient and target information is imported into Cytoscape 3.7.1 and all the information was entered into the software as nodes, which were related to each other by line segments. The software measured these nodes according to an algorithm and output a degree value for each node (Hu et al., 2022). We can sort the nodes according to the degree value, and then highlight the important nodes by changing the color depth and node area size.

2.2.3. Screening for disease genes

We found the disease genes that were associated with acute kidney injury after severe burns in three disease gene databases, GeneCards (<https://www.genecards.org>), DrugBank (<https://go.drugbank.com>), DisGeNET (<https://www.disgenet.org>).

2.2.4. Predicting targets that play an important role

We intersected the targets acted on by HQJND with disease genes and then visualized such intersections as a Wayne diagram (Bardou et al., 2014).

2.2.5. Enrichment analysis of critical targets

We performed Gene ontology (GO) and Kyoto encyclopedia of genes and genomes (KEGG) enrichment analysis and visualization of target targets obtained in 2.2.4. by using R packages such as clusterProfiler, enrichplot and ggplot2.

2.2.6. Establishing protein-protein interaction (PPI) networks

We imported the genes obtained in 2.2.4. into the string database (<https://string-db.org/>) to analyze functional relationships between protein sequences of species. The information on protein relationships obtained from string was imported into cytoscape, which distinguished and visualized protein nodes based on the information from string.

2.2.7. Molecular docking

Molecular docking is a computerized technique to simulate the binding of a receptor to a ligand (Ferreira et al., 2015). We downloaded the X-ray structures of the target proteins from the Protein Structure Database (<https://www.rcsb.org>), and then we downloaded the structures of some small molecules that played a major role in the Chinese

medicine mixtures from PubChem (<https://pubchem.ncbi.nlm.nih.gov>). Next, we simulated the receptor-ligand binding and measured the binding and interaction relationships between them in AutoDockTools and finally we visualized the results in pymol.

2.3. Animal experiment

2.3.1. Establishing animal model

Healthy male Sprague-Dawley rats (200–250 g, 8–10 weeks) were provided by Jinan Pangyue Experimental Animal Breeding Co Ltd (Approval No. 2019-0003). This study was approved by the Experimental Animal Ethics Committee of Anhui Medical University (LLSC20190251). The rats were acclimatized and fed for one week and then grouped for treatment. Twenty-four rats were randomly divided into three groups, namely, the sham-operated group, the model group, and the herb-administered group, with eight rats in each group. After anesthetized with isoflurane, all rats were shaved. Then, the backs of the rats in the sham-operated group were moistened with warm water at 37 °C for 15s while the backs of the rats in the model group as well as the rats in the herb-administered group were exposed to boiling water at 98 °C for 15s. Next, the backs of the rats in the three groups were covered with clean gauze to absorb the excess water, and were disinfected with iodine and the gauze was covered with adhesive tapes to fix the skin. In addition, 2 mg/kg of Lipopolysaccharide (LPS) was injected intraperitoneally into rats in the model group and the herb-delivered group to create a rat model with 30% of the burned area, a depth of III degree, and rats with sepsis were combined. After successful modeling, gastric gavage was started. The sham-operated group and the model group were gavaged with an equal volume of distilled water, and the herb administration group was gavaged with a dose of 1 ml of the concentrated herb twice a day for 3 days. The dosage was according to the conversion of the body surface area of the rats to that of the humans.

2.3.2. Sample collection

At 1 h after the last gavage treatment, the rats were anesthetized and placed in supine position for cardiac blood sampling. About 4 ml of blood could be taken from each rat and allowed to stand for 4 h in a refrigerator at 4 °C. Then the blood was centrifuged in a freezing centrifuge at a speed of 3000 rpm at 4 °C for 10 min, and the serum was taken and stored in a refrigerator at –80 °C. The abdominal cavity of rats was opened along the ventral midline to expose the kidneys, and the kidneys were removed on ice to peel off the surface envelope. One part of the kidneys was submerged in fixative solution for subsequent pathologic testing, and the other part was packed into freezing tubes and stored in a –80 °C refrigerator.

2.3.3. Kidney function test

Serum levels of creatinine and urea nitrogen in the samples taken were measured using a fully automated biochemistry (mindray, BS-240Vet).

2.3.4. Renal pathology

First, rat kidneys were fixed in the 4% paraformaldehyde for 24 h, followed immediately by dehydration, paraffin embedding, and sectioning. Then, the paraffin sections were deparaffinized with xylene and different concentrations of gradient ethanol. The sections were stained using hematoxylin and eosin staining solution. Afterwards, the stained sections were in transparent processing with different concentrations of ethanol and xylene. Finally, the sections were sealed with neutral gum. We observed the degree of renal injury using a microscope and the observations included brush border loss, tubular nuclei loss, hemorrhage, intraepithelial vacuolar degeneration, diffusely dilated kidney tubules and intratubular cast. We scored the degrees by assessing

the percentage of renal injury area on a scale of 0: no injury; 1: 1%–25%; 2: 26%–50%; 3: 51%–75%; 4: 76%–100% (Yan et al., 2020). For renal immunohistochemistry testing, we boiled the dewaxed sections in citrate buffer for antigen repair. After the sections were cooled, peroxidase was eliminated with 3% H₂O₂ and the tissues were then closed with goat serum. After closure, they were incubated with Kidney injury molecule 1 (KIM1) antibody for nights. The secondary antibodies was incubated in a 37 °C thermostat for 30 min, then 3,3-N-Diaminobenzidine Tetrahydrochloride (DAB) was added to develop the color, and hematoxylin was used to restrain the color after DAB development. Finally, the sections was dehydrated with gradient alcohol, transparent with xylene and sealed with neutral gum. For immunohistochemistry results, we used ImageJ to analyze the percentage of positive area.

2.3.5. Western Blotting

Kidney cortical tissues were taken, and the tissues were ground and lysed on ice with RIPA lysate, phenyl methane sulfonyl fluoride, protease inhibitor and phosphatase inhibitor to obtain total proteins (Hu et al., 2023). Different molecular weight proteins in the total proteins were separated using sds-page gel electrophoresis, and then the proteins on the gel were transferred to Nitrocellulose membranes. The proteins on the membranes were blocked by Bovine Serum Albumin for 1.5 h, and then incubated with primary antibodies (Phospho-Nuclear factor- κ B p65 (PP65), Cell Signaling Technology, 1:1000; Nuclear factor- κ B p65 (P65), Cell Signaling Technology, 1:1000; NF- κ B inhibitor α (I κ B- α), proteintech, 1:4000; phospho-NF- κ B inhibitor α (P-I κ B- α), bioworlde, 1:1000; Glyceraldehyde-3-phosphate dehydrogenase (GAPDH), proteintech, 1:5000; Tumor necrosis factor receptor superfamily member 1A (TNFR1), proteintech, 1:500; TNF α , Abcam, 1:1000; KIM1, BIOSS, 1:2000.) overnight in a 4 °C refrigerator. After incubation of the primary antibody, the membrane was washed three times with Tris-Buffered Saline Tween-20 (TBST) for 10–15 min each time. The secondary antibody (horseradish enzyme labeled goat anti-rabbit IgG, ZSGB-BIO, 1:5000) and membranes were incubated at room temperature for 1 h and then washed with TBST. Finally, the target proteins on the Nitrocellulose membrane were developed using exposure solution (Beyotime) and the target bands were analyzed by ImageJ.

2.3.6. Quantitative real-time PCR

RNA from kidney tissues was first extracted on ice using a kit (YiShan Biotech). Next, after RNA concentration measurement, the RNA was reversely transcribed into cDNA using a kit (TaKaRa, RR047A, RR820A), and then amplified and fluorescently quantified using the cDNA as a template under a system (Applied Biosystems 7500). The primer sequences are shown in Table 2.

2.4. Statistical methods

GraphPad Prism 9 was used for statistical analysis. One-way Analysis of Variance (ANOVA) was used to compare differences between groups. The experimental data in the graphs were expressed as mean $\pm$ standard error of mean. $p < 0.05$, the difference was statistically significant.

3. Results

3.1. Identification of HQJND components by LC-MS/MS

The contents of Geniposidic acid, Paeoniflorin, Caffeic acid, Lobe-tyolin, Ferulic Acid, Azelaic Acid, Formononetin, Atractylenolide III, Astragaloside IV, Tanshinone I, Tanshinone IIA and Schisandrin B were 0.068, 2.28, 0.018, 0.204, 0.762, 0.166, 0.019, 0.133, 1.128, 0.338, 0.00012 and 0.0034 mg/g in HQJND, respectively (Table 3) (see Fig. 1).

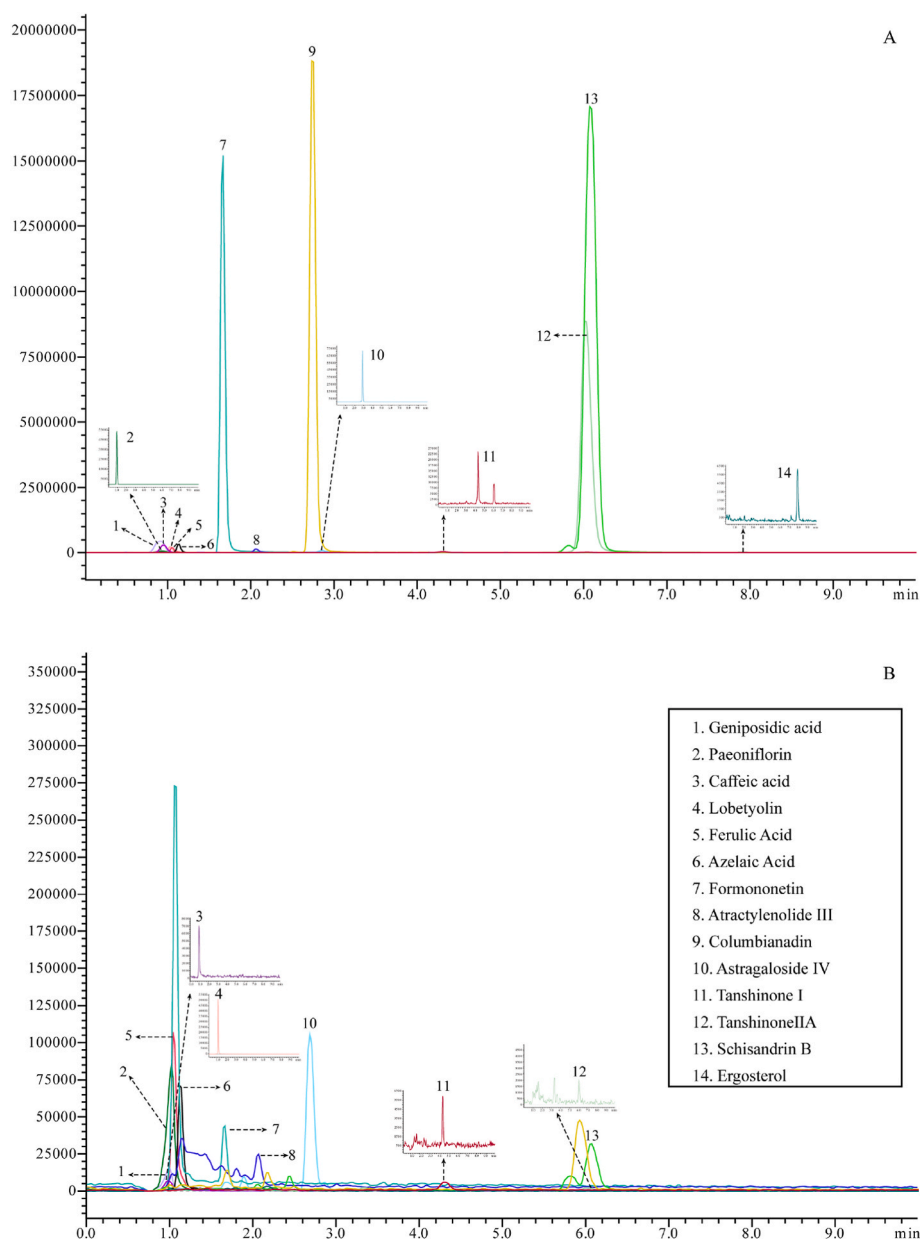


Fig. 1. Typical total ion multiple-reaction monitoring chromatograms of 14 mixed standard compounds (A) and HQJND (B).

3.2. Network pharmacology

3.2.1. Drug target information

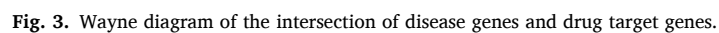
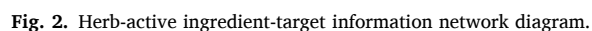
We collected information on targets that were closely related to HQJND through two databases, PharmMapper and TCMSP. It was found that the 12 Chinese medicines was comprised of a total of 139 main components, which could act on 286 protein targets. The relationship between drugs, active ingredients and targets was shown in Fig. 2. The middle rhombus is the target gene of the drug and if the color was darker, the larger area of the rhombus node represented that the gene was the target of multiple active ingredients of the drug. The nodes around the rhombus represented the drugs and their active ingredients. If the drug had active ingredients, the drug had larger area and darker color. Similarly, if the active ingredient had more acting targets, it had a larger area and a darker color, such as the more conspicuous nodes: E01: quercetin, A02: kaempferol, G01: luteolin, F01: Stigmasterol, B01: beta-sitosterol, and so on.

3.2.2. PPI networks of intersecting genes in diseases and drugs

A total of 353 genes closely related to diseases were collected from three disease databases, GeneCard, Disgenet and OMIM. There were 32 common genes between these 353 genes and 286 targets in HQJND. The interactions between these 32 targets were shown in Fig. 3 which demonstrated that inflammation-related targets took an important position.

3.2.3. KEGG, GO enrichment analysis

The 32 target genes were visualized by enrichment analysis of the top 20 signaling pathways, which were closely related to 22 genes, Interleukin-6 (IL6), TNF, Interleukin-1 beta (IL1B), Interleukin-2 (IL2), C-X-C motif chemokine ligand 8 (CXCL8), Interleukin-10 (IL10), Matrix metalloproteinase-2 (MMP2), Fos proto-oncogene (FOS), Pro-epidermal growth factor (EGF), C-X-C motif chemokine 2 (CXCL2), Interferon gamma (IFNG), Interleukin-4 (IL4), Vascular cell adhesion molecule-1 (VCAM1), B-cell lymphoma 2 (BCL2), Caspase-3 (CASP3), Caveolin-1 (CAV1), Collagen alpha-1(III) chain (COL3A1), Interferon regulatory factor 1 (IRF1), BCL2



associated X (BAX), Heme oxygenase 1 (HMOX1), Signal transducer and activator of transcription 3 (STAT3). The 22 genes were the targets acted on by *Rehmannia glutinosa* (Gaertn.) DC., *Astragalus mongholicus* Bunge, *Angelica sinensis* (Oliv.) Diels, *Plantago ovata* Forssk., *Atractylodes macrocephala* Koidz., *Salvia miltiorrhiza* Bunge, *Codonopsis pilosula* (Franch.) Nannf. and *Paeonia lactiflora* Pall., so we believed that these eight herbs exerted a major efficacy in HQJND (Fig. 4E). Based on the results of gene

enrichment (NF- κ B signaling pathway, TNF signaling pathway, mononuclear cell differentiation, leukocyte cell-cell adhesion, mononuclear cell differentiation, leukocyte cell-cell adhesion, lymphocyte differentiation, regulation of T cell activation), it is obvious that the mechanism of HQJND treatment was closely related to inflammation. In addition, the NF- κ B signaling pathway was interconnected with the TNF signaling pathway and was inextricably linked to the development of

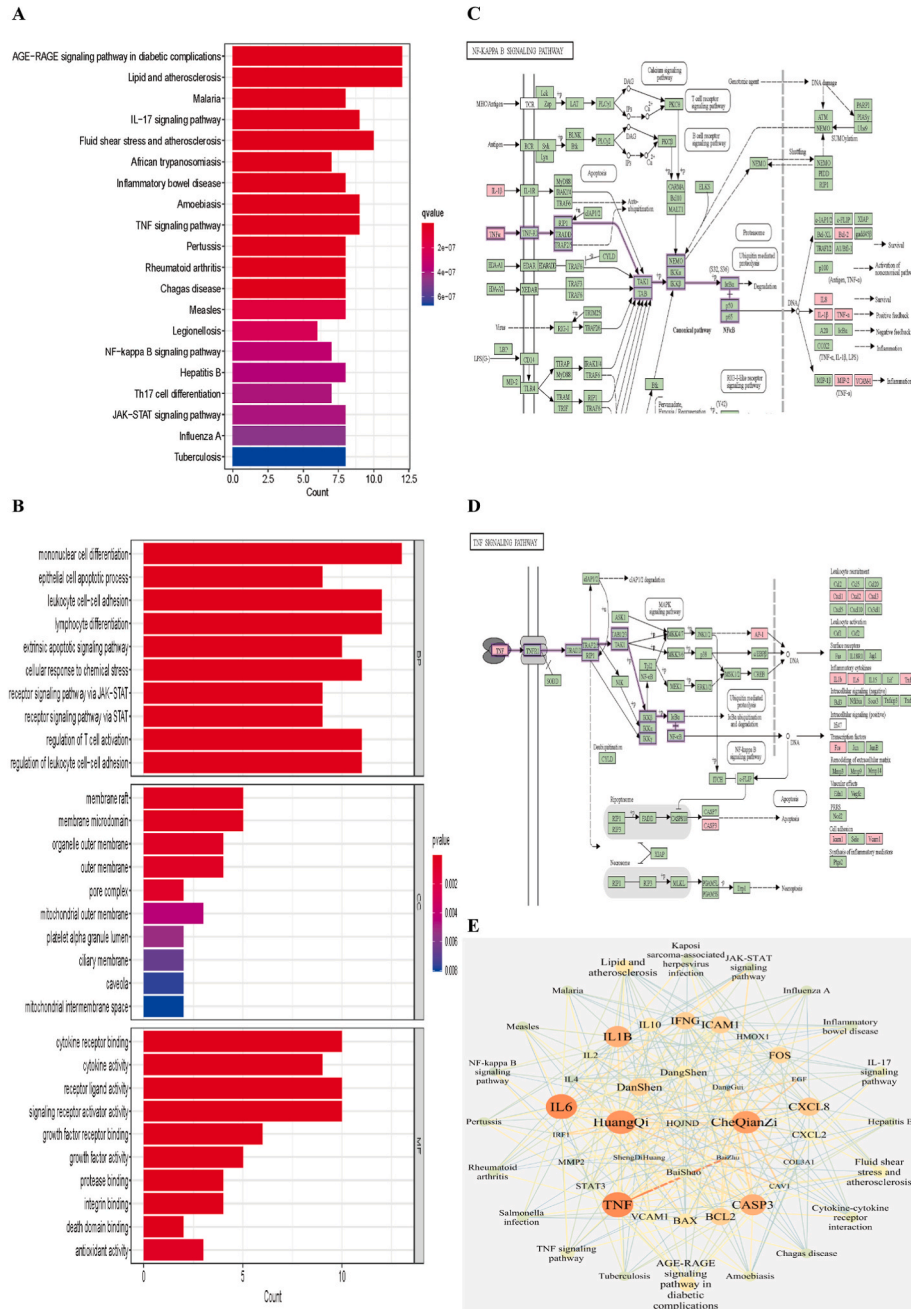


Fig. 4. (A.C.D.) KEGG pathway analysis. (B) GO analysis. (E) Top 20 signaling pathways obtained from KEGG pathway analysis, their respective genes involved, and drug-drug associations.

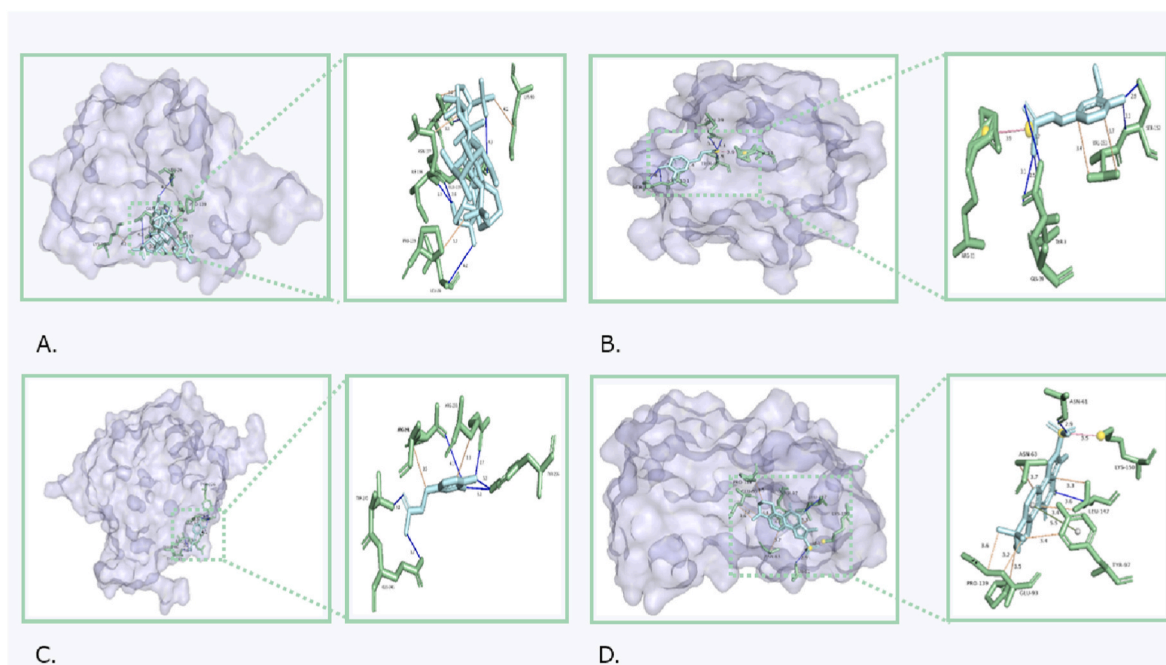


Fig. 5. Plot of molecular docking results of the main chemical components of HQJND with the target targets. (A) Astragaloside IV-TNF α ; (B) Ferulic acid-IL-1 β ; (C) Caffeic acid-Casp3; (D) Tanshinone IIA -IL-6.

inflammation (Liu, 2005), so we chose to explore in experiments how HQJND could exert therapeutic effects through the TNF/NF- κ B signaling pathway.

3.2.4. Molecular docking

We combined the results of network pharmacology with TCM pharmacological knowledge (Garba et al., 2020) and screened four monomers that played crucial roles in the formulas: Astragaloside IV, Ferulic Acid, Tanshinone IIA, Caffeic Acid as ligands, and target proteins that occupied an important position in the PPI network: IL-1 β , Casp3, TNF α , IL-6 as receptors. The receptor-ligand interactions were shown in Fig. 5, and the binding energies of action were shown in Table 4.

3.3. Animal experiment

3.3.1. The effect of renal function in rats and histopathologic intervention

HQJND has a therapeutic effect on rats with severe burns combined with sepsis. Hematoxylin and eosin staining of rat kidneys revealed less renal tubular injury in the sections of rats Sham and Herb groups, while scattered hemorrhagic spots in renal tissues could be seen in the kidney sections of rats in the Model group. Not only that, in the Model group, the presence of tubular pattern with enlarged lumen and loss of brush border could be seen in the renal tubules and the cells were in the form of vacuoles, and the nuclei of the cells were detached to the lumen. HE renal tubular injury score showed that the degree of renal injury in the rats in the Herb group was significantly lower than that in the Model and Sham-operated groups. From serum renal function to IHC, WB, RT-qPCR of renal tissues in rats, we could find that markers representing renal tissue injury were highly expressed in the Model and suppressed in the Herb-treated group (see Fig. 6), which demonstrated the good therapeutic efficacy of our herbal combination.

3.3.2. Effect of HQJND on TNF/NF- κ B signaling pathway

TNF and NF- κ B signaling pathways have been shown to be associated with acute kidney injury (Shu et al., 2019; Zhao et al., 2022). Together

with the results of network pharmacology prediction, we validated several key targets of TNF α , TNFR1, PP65, P65, PIK β , and IK β in the TNF/NF- κ B pathway by WB, RT-qPCR. Normally, IK β binds to P65 and inhibits gene transcription caused by nuclear translocation of P65. However, when TNFR1 recognizes TNF α , it will result in phosphorylation of IK β and thus cannot continue to isolate P65 in the cytoplasm, which causes P65 to translocate to the nucleus and the whole signaling pathway is activated, triggering uncontrolled inflammation causing tissue damage (Oeckinghaus et al., 2011). The results showed that the expressions of TNF α , TNFR1, PP65, P65, and PIK β proteins in the kidneys of rats in the Model group was significantly higher than that in the Sham and Herb groups, while IK β was higher in the Sham and Herb groups than that in the Model group. At the transcript level, mRNA levels of TNF α , TNFR1, and P65 followed the same trend as protein expression (see Fig. 7). This indicated that the TNF/NF- κ B pathway was inhibited after drug treatment, which in turn alleviated the inflammatory response of the kidney and reduced the tissue damage.

4. Discussion

Acute kidney injury is one of the possible complications of severe burns. It should be given great attention because of its high mortality rate, long hospitalization time, high treatment cost and possible accompanying sequelae (Palmieri et al., 2010). Burn-induced renal injury is categorized into early renal injury (within one week after injury) and late renal injury (one week after injury) (Witkowski et al., 2016). When severe burns first occur, a large amount of body fluids are lost from the trauma. Moreover, vascular permeability increases while circulating blood volume decreases, and ischemia and hypoxia occur in the tissues. At this time, to guarantee the blood supply to the heart and the brain, the blood is redistributed to, which will further aggravate the acute injury of renal tissues (John, 2020). Whereas, sepsis as well as multiorgan failure is the cause of late stage of acute kidney injury (Clark et al., 2017). However, whether it is early or advanced AKI, its onset and development and prognosis are inextricably linked to inflammation

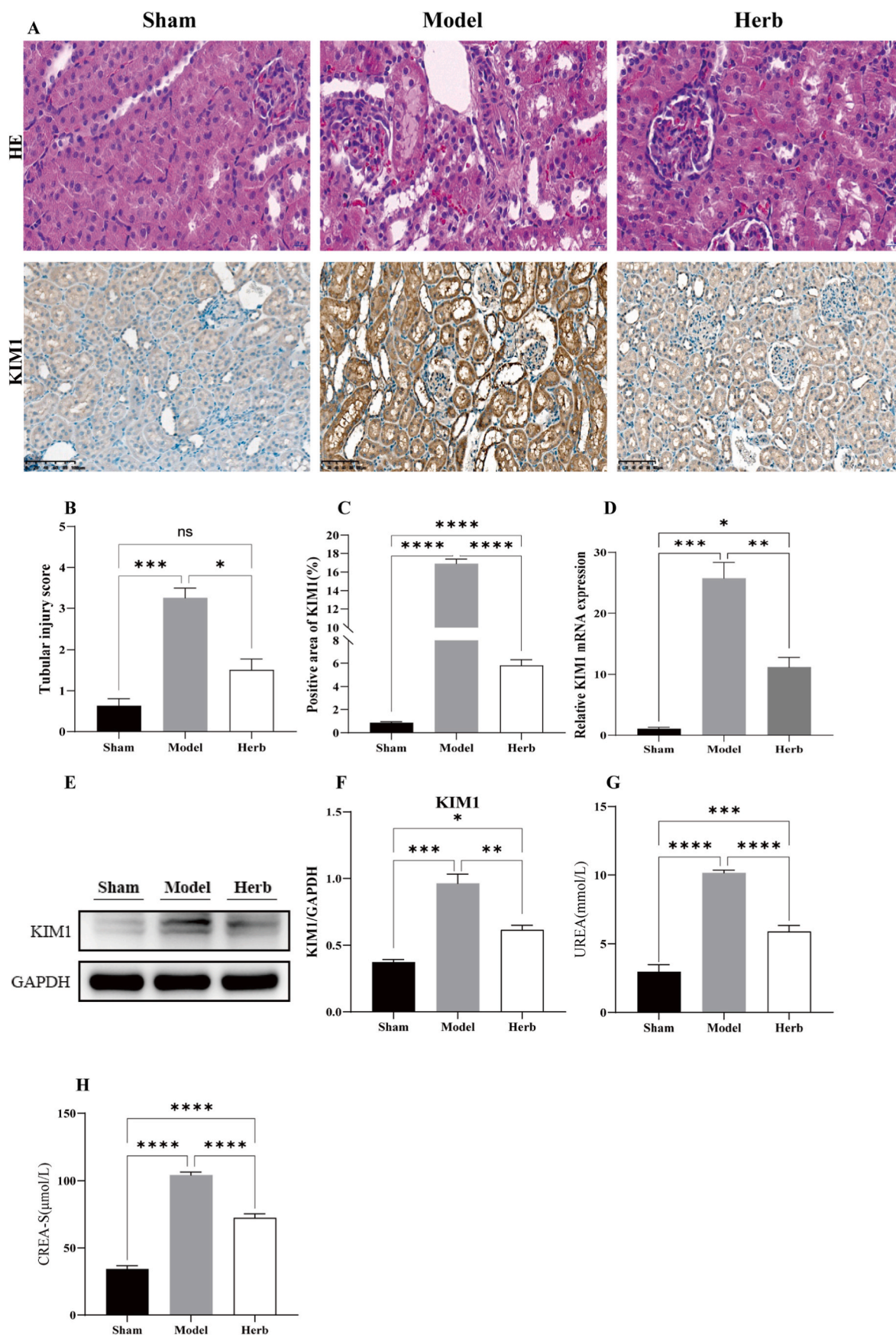


Fig. 6. Effects of HQJND on acute kidney injury in rats. (A) Kidney sections of HE and IHC rats; (B) Analysis of the extent of kidney injury in HE sections of rats; (C) ImageJ analysis of positive area in IHC sections; (D–F) Protein and mRNA expression levels of the kidney injury marker KIM1; (G–H) Serum renal function in rats. Data are expressed as SD ± SEM, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Table 4
Binding energy of compounds and core targets.

Receptor	PDB ID	Ligand	Binding energy (kcal/mol)
IL-1 β	5r7w	Astragaloside IV	−5.8
		Ferulic acid	−6.06
		Tanshinone IIA	−9.07
		Caffeic Acid	−6.37
Casp3	5ibp	Astragaloside IV	−5.25
		Ferulic acid	−5.84
		Tanshinone IIA	−7.89
		Caffeic Acid	−5.67
TNF α	5uui	Astragaloside IV	−6.07
		Ferulic acid	−6.33
		Tanshinone IIA	−8.25
		Caffeic Acid	−5.46
IL-6	1alu	Astragaloside IV	−5.61
		Ferulic acid	−5.28
		Tanshinone IIA	−8.43
		Caffeic Acid	−6.25

(Rabb et al., 2016).

Traditional Chinese medicines has unique advantages in treating AKI, because they can alleviate renal injury through different mechanisms (Li, H.D. et al., 2019). Modern pharmacological studies have shown that each herb in HQJND have renal protective effects. For example, *Astragalus mongholicus* Bunge reduces sepsis-induced renal tubular injury by regulating the PI3K/AKT pathway (Tang et al., 2022). *Codonopsis pilosula* (Franch.) Nannf can regulate the water-electrolyte balance in the body (Chen and Wu, 2021). *Atractylodes macrocephala* Koidz. has diuretic, edema-eliminating, anti-inflammatory and antioxidant effects (Yang et al., 2021). *Angelica sinensis* (Oliv.) Diels not only promotes the proliferation of renal tubular epithelial cells, but also inhibits its apoptosis (Bunel et al., 2015). *Rehmannia glutinosa* (Gaertn.) DC delayed the progression of diabetic nephropathy (Yokozawa et al., 2004). *Paeonia lactiflora* Pall. hinders chemotherapeutic drug-induced renal injury because of the regulation of the AMPK/NF- κ B pathway (Liu et al., 2016). The active ingredient of *Salvia miltiorrhiza* Bunge can inhibit the release of inflammatory factors to treat AKI (Zeng et al., 2020). *Schisandra chinensis* Turcz. (Baill.) can inhibit oxidative stress, inflammation, and apoptosis (Li et al., 2018). The active ingredient in *Alisma plantago-aquatica* subsp. *orientale* (Sam.) Sam can improve renal function by activating nuclear receptors and then regulating gene transcription (Luan et al., 2021). The treatment of gout-induced renal injury by *Plantago ovata* Forssk. is achieved through the inhibition of inflammatory signaling pathways (Zhao et al., 2021). *Polyporus umbellatus* (Pers.) Fries can treat renal interstitial fibrosis through a variety of complex mechanisms (Li, H. et al., 2019). The protective properties of *Poria cocos* (Schw.) Wolf for the kidneys are reflected in its ability to increase urine output and then excrete excess uric acid, thus preventing hyperuricemic hypercalcemia (Liang et al., 2021). Chinese medicines usually work together with multiple active ingredients at multiple targets when protecting the body from excessive inflammatory responses (Wang et al., 2013). Network pharmacology validation indicated that HQJND treatment also involves multiple targets and signaling pathways. So, we selected six key targets in the TNF/NF- κ B pathway for experimental validation. Both the TNF signaling pathway and the NF- κ B signaling pathway are inflammation-related pathways (Dostert et al., 2019; Sarkar et al., 2008), and the NF- κ B pathway can be activated by the TNF receptor family (Walker et al., 2019). In addition, studies have argued that the TNF/NF- κ B signaling pathway correlated with dysfunction and injury of renal endothelial cells and renal proximal tubular epithelial cells (Han et al., 2022; Wu et al., 2009). TNF α is a pro-inflammatory cytokine produced by immune cells and induced by

LPS (Yoshimura et al., 2003), and its cytotoxicity and pro-inflammatory effects can be attenuated by knockout of the gene in mice with AKI (Ramesh and Reeves, 2002). TNFR1 is a cell membrane receptor found in renal tubular and glomerular endothelial cells, which has been implicated in renal immunity, inflammation, tumors, and injury (Al-Lamki and Mayadas, 2015). IKB α , also known as NF κ B inhibitor α , binds to and inhibits NF κ B in the cytoplasm; when cells are stimulated by, for example, inflammation, IKB α protein is phosphorylated to PIK β , which allows NF κ B to undergo nuclear translocation (Israel, 2010). When polyubiquitination and proteasomal degradation of proteins of the NF κ B inhibitor family occurs, P65, a component of NF κ B, is released and translocated to the nucleus to regulate gene transcription (Cartwright et al., 2016). PP65, which is phosphorylated by a variety of kinases, potentiates NF κ B transcriptional activity (Liu et al., 2008). We observed a significant reduction in the expression levels of TNF α , TNFR1, PP65, P65 and PIK β , and a significant increase in IKB α content in rat kidney after HQJND administration. This implies that HQJND is reducing the inflammatory level in rat kidney by inhibiting the activation of TNF/NF- κ B pathway (Fig. 8). After the out-of-control inflammatory response in the kidney was controlled, AKI was also alleviated, which could be observed in the reduced expression of the kidney injury marker-KIM1, the reduction of serum creatinine and urea nitrogen as well as the reduced degree of tubular injury observed in pathological sections.

With the research advances, people admit that the pathogenesis of diseases is complex and that the control of disease progression with single-target drugs is difficult (Noor et al., 2022). Meanwhile, Chinese medicines are unique in holistic concept of drug administration, are rich in active ingredients and a wide range of therapeutic targets, so it may probably be a promising direction to apply traditional Chinese medicine in the treatment of diseases (Li and Zhang, 2013). Therefore, the advantage of our decoction is that it has better efficacy and wider therapeutic range compared to western drugs and herb monomers. For example, HIF-1 α reduces renal injury by inhibiting vascular permeability (Jiang et al., 2021); lutein exerts antioxidant, anti-inflammatory, and anti-apoptotic effects to protect the kidneys (AbuBakr et al., 2018) and the network pharmacological predictions of HQJND show that it acts on many signaling pathways and participates in a wide range of biological processes, which is efficacious in treating the disease at multiple targets. In addition, HQJND not only have multiple targets but also have synergistic effects between various drugs, which can substantially reduce the dosage of each drug compared with single drugs, thereby reducing drug toxicity and enhancing efficacy (Nogales et al., 2022). In addition to the therapeutic advantages, this study is advantageous in research methodology. The research model of network pharmacology is "multi-target, multi-mechanism", so it is extremely suitable for the study of herbal compound, and can help researchers to understand the drug more comprehensively and find the correct research directions (Zhang et al., 2019). In addition, compared with the exploration of drug efficacy (Yuan et al., 2019), we also explored the molecular mechanism of HQJND, and testified how the drug could reduce kidney injury in rats through the TNF/NF- κ B pathway.

Many of the active ingredients in HQJND have been shown to be effective in anti-inflammation as far as LC-MS/MS results are concerned: geniposidic acid attenuated inflammatory bowel disease in mice through modulation of the NF κ B pathway (Jiang et al., 2023). Astragaloside IV, Paeoniflorin and Tanshinone IIA treated Alzheimer's disease through the NF- κ B pathway (He et al., 2023; Seo et al., 2018). Modulation of the NF- κ B signaling pathway by Caffeic acid and Ferulic Acid treated inflammatory diseases of the nervous system (Kwon et al., 2019). Formononetin attenuated gastric ulcers in rats by altering protein

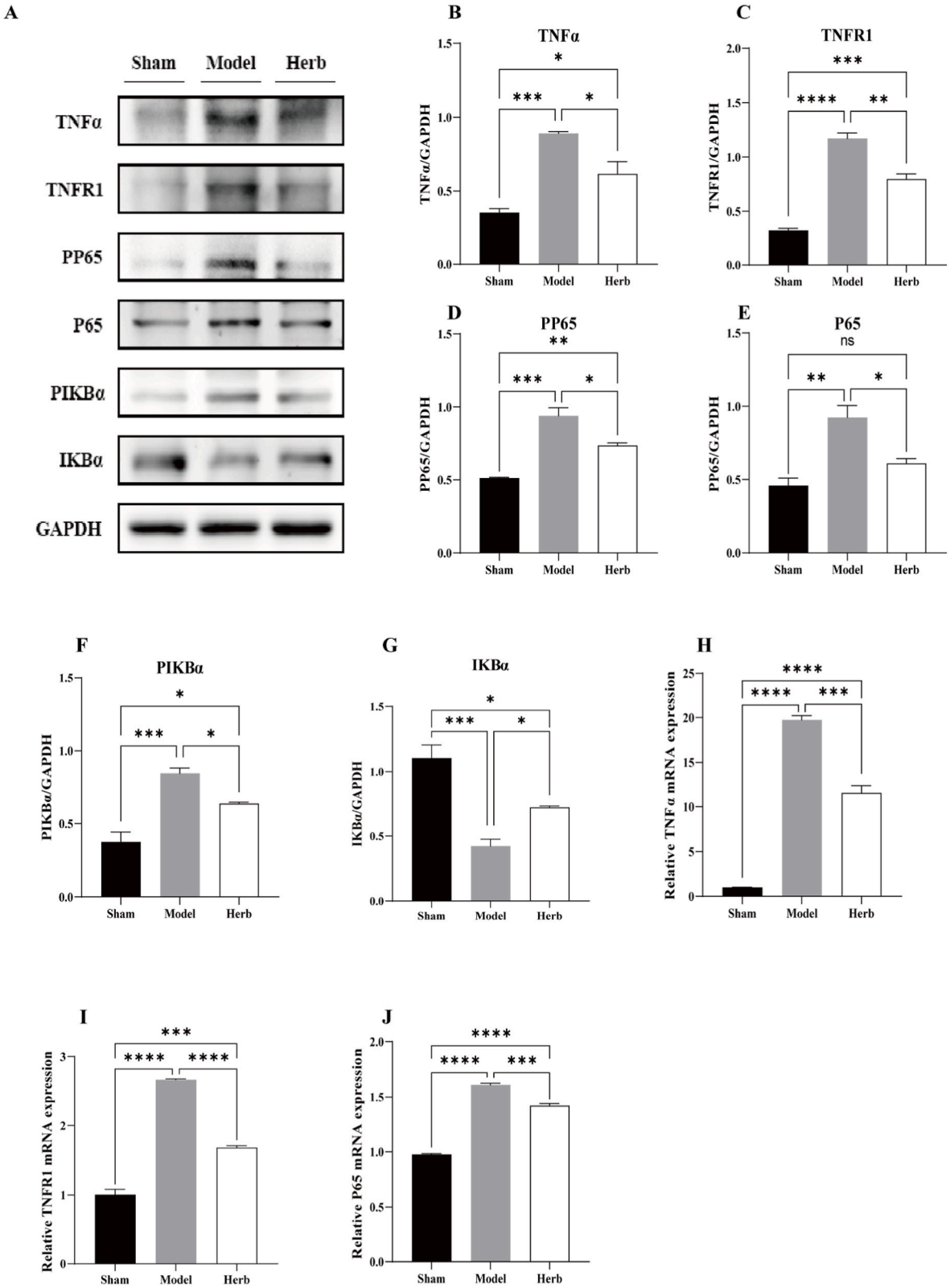


Fig. 7. HQJND inhibits the TNF/NF- κ B signaling pathway in AKI rats. (A–G) Protein blotting of TNF α , TNFR1, PP65, P65, PIKBa, IKBa proteins in the kidney and their quantitative analysis (n = 3); (H–J) TNF α , TNFR1, P65 mRNA levels by qRT-PCR.

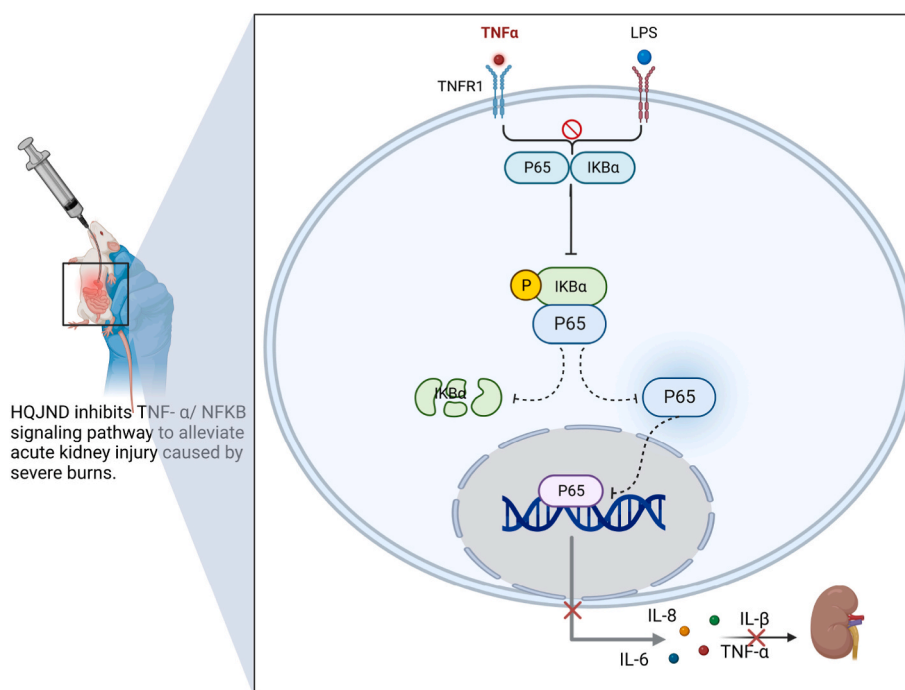


Fig. 8. Diagram of TNF/NF- κ B signaling pathway inhibition by HQJND(Created with BioRender.com).

expression of the NF- κ B pathway (Yi et al., 2022). Columbianadin, Schisandrin B, and Atractylenolide III all alleviated osteoarthritis through the NF- κ B signaling pathway (Chen et al., 2023; Ran et al., 2018; Xu et al., 2023). Ergosterol blocked NF- κ B signaling pathway activation, thereby inhibiting renal inflammatory response in mice with diabetic nephropathy (Liu et al., 2020). In addition, both network pharmacology results and experimental results verified that HQJND attenuates acute kidney injury in rats by modulating the TNF/NF- κ B signaling pathway.

5. Conclusion

Network pharmacology was utilized to excavate the mechanism of acute kidney injury induced by HQJND for the treatment of severe burns. It was found that it may act through the TNF/NF- κ B pathway. Following this, we went on to successfully validate this in rats. Therefore, we concluded that HQJND attenuates acute kidney injury by inhibiting the activation of the TNF/NF- κ B signaling pathway.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Funding

This study was sponsored by the National Natural Science Fund (NO:81,903,994), Youth Development of the First Affiliated Hospital of Anhui Medical University (NO:2793), the Key projects of the clinical research of the First Affiliated Hospital of Anhui Medical University (NO: LCYJ2021ZD005), Anhui Medical University College Students Innovation and Entrepreneurship Project (Provincial Project, 2022).

CRediT authorship contribution statement

Wanxuan Hu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Jie Zhao:** Conceptualization, Writing – review & editing, Supervision, Project administration, Data curation, Funding acquisition. **Yuxin Hu:** Formal analysis. **Shuai Song:** Visualization. **Xulin Chen:** Writing – review & editing. **Yexiang Sun:** Conceptualization, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.117344>.

References

- AbuBakr, H.O., Aljuaydi, S.H., Abou-Zeid, S.M., El-Bahrawy, A., 2018. Burn-induced multiple organ injury and protective effect of lutein in rats. *Inflammation* 41 (3), 760–772. <https://doi.org/10.1007/s10753-018-0730-x>.
- Al-Lamki, R.S., Mayadas, T.N., 2015. TNF receptors: signaling pathways and contribution to renal dysfunction. *Kidney Int.* 87 (2), 281–296. <https://doi.org/10.1038/ki.2014.285>.
- Bagshaw, S.M., Darmon, M., Ostermann, M., Finkelstein, F.O., Wald, R., Tolwani, A.J., Goldstein, S.L., Gattas, D.J., Uchino, S., Hoste, E.A., Gaudry, S., 2017. Current state of the art for renal replacement therapy in critically ill patients with acute kidney injury. *Intensive Care Med.* 43 (6), 841–854. <https://doi.org/10.1007/s00134-017-4762-8>.
- Bardou, P., Mariette, J., Escudie, F., Djemiel, C., Klopp, C., 2014. jvenn: an interactive Venn diagram viewer. *BMC Bioinf.* 15, 7. <https://doi.org/10.1186/1471-2105-15-293>.
- Beccaria, M., Cabooter, D., 2020. Current developments in LC-MS for pharmaceutical analysis. *Analyst* 145 (4), 1129–1157. <https://doi.org/10.1039/c9an02145k>.

- Bunel, V., Antoine, M.H., Nortier, J., Duez, P., Stévigny, C., 2015. Potential nephroprotective effects of the Chinese herb *Angelica sinensis* against cisplatin tubulotoxicity. *Pharm. Biol.* 53 (7), 985–994. <https://doi.org/10.3109/13880209.2014.951726>.
- Cartwright, T., Perkins, N.D., Wilson, C.L., 2016. NFKB1: a suppressor of inflammation, ageing and cancer. *FEBS J.* 283 (10), 1812–1822. <https://doi.org/10.1111/febs.13627>.
- Chen, S., Wang, Y., Zhang, L., Han, Y., Liang, C., Wang, S., Qi, L., Pang, X., Li, J., Chang, Y., 2023. Therapeutic effects of columbinadin from *Angelica Pubescentis* radix on the progression of collagen-induced rheumatoid arthritis by regulating inflammation and oxidative stress. *J. Ethnopharmacol.* 316, 116727. <https://doi.org/10.1016/j.jep.2023.116727>.
- Chen, S., Wu, X., 2021. Codonopsis Radix modulates water and electrolytes homeostasis in mice. *Heliyon* 7 (4), e06735. <https://doi.org/10.1016/j.heliyon.2021.e06735>.
- Clark, A., Neyra, J.A., Madni, T., Imran, J., Phelan, H., Arnoldo, B., Wolf, S.E., 2017. Acute kidney injury after burn. *Burns* 43 (5), 898–908. <https://doi.org/10.1016/j.burns.2017.01.023>.
- Crampon, K., Giorkallos, A., Deldossi, M., Baud, S., Steffanel, L.A., 2022. Machine-learning methods for ligand-protein molecular docking. *Drug Discov. Today* 27 (1), 151–164. <https://doi.org/10.1016/j.drudis.2021.09.007>.
- Dostert, C., Grusdat, M., Letellier, E., Brenner, D., 2019. The TNF family of ligands and receptors: communication modules in the immune system and beyond. *Physiol. Rev.* 99 (1), 115–160. <https://doi.org/10.1152/physrev.00045.2017>.
- Ferreira, L.G., dos Santos, R.N., Oliva, G., Andricopulo, A.D., 2015. Molecular docking and structure-based drug design strategies. *Molecules* 20 (7), 13384–13421. <https://doi.org/10.3390/molecules200713384>.
- Folkestad, T., Brurberg, K.G., Nordhuus, K.M., Tveiten, C.K., Guttormsen, A.B., Os, I., Beitland, S., 2020. Acute kidney injury in burn patients admitted to the intensive care unit: a systematic review and meta-analysis. *Crit. Care* 24 (1), 11. <https://doi.org/10.1186/s13054-019-2710-4>.
- Garba, Z.N., Zhou, W.M., Zhang, M.X., Yuan, Z.H., 2020. A review on the preparation, characterization and potential application of perovskites as adsorbents for wastewater treatment. *Chemosphere* 244, 23. <https://doi.org/10.1016/j.chemosphere.2019.125474>.
- Han, C., Zheng, J., Wang, F.Y., Lu, Q.Y., Chen, Q.F., Hu, A.K., Visentin, M., Kullak-Ublick, G.A., Gai, Z.B., Chu, L., 2022. The role of NF- κ B in the downregulation of organic cation transporter 2 expression and renal cation secretion in kidney disease. *Front. Med.* 8, 12. <https://doi.org/10.3389/fmed.2021.800421>.
- He, L., Sun, J., Miao, Z., Chen, S., Yang, G., 2023. Astragaloside IV attenuates neuroinflammation and ameliorates cognitive impairment in Alzheimer's disease via inhibiting NF- κ B signaling pathway. *Heliyon* 9 (2), e13411. <https://doi.org/10.1016/j.heliyon.2023.e13411>.
- Hu, X., Pan, G., Luo, J., Gao, X., Mu, Y., Wang, Z., Hu, X., Li, C., Abbas, M.N., Zhang, K., Zheng, Y., Cui, H., 2023. Kuwanon H inhibits melanoma growth through cytotoxic endoplasmic reticulum stress and impaired autophagy flux. *J. Agric. Food Chem.* 71 (37), 13768–13782. <https://doi.org/10.1021/acs.jafc.3c02257>.
- Hu, X., Zhang, K., Pan, G., Wang, Y., Shen, Y., Peng, C., Deng, L., Cui, H., 2022. Cortex Mori extracts induce apoptosis and inhibit tumor invasion via blockage of the PI3K/AKT signaling in melanoma cells. *Front. Pharmacol.* 13, 1007279. <https://doi.org/10.3389/fphar.2022.1007279>.
- Hui, Z., Dong, Q.Q., Shu, H.P., Tu, Y.C., Liao, Q.Q., Yao, L.J., 2021. Mechanistic insights into the renoprotective role of curcumin in cisplatin-induced acute kidney injury: network pharmacology analysis and experimental validation. *Bioengineered* 12 (2), 11041–11056. <https://doi.org/10.1080/21655979.2021.2005916>.
- Israel, A., 2010. The IKK complex, a central regulator of NF- κ B activation. *Cold Spring Harbor Perspect. Biol.* 2 (3), 14. <https://doi.org/10.1101/cshperspect.a000158>.
- Jiang, P., Zhang, Y., Li, X., Chen, J., 2023. Geniposidic acid attenuates DSS-induced colitis through inhibiting inflammation and regulating gut microbiota. *Phytother. Res.* 37 (8), 3453–3466. <https://doi.org/10.1002/ptr.7819>.
- Jiang, W., Sun, Y., Wang, H., Hu, Z., Song, J., Meng, C., Duan, S., Jiang, Z., Yu, Y., Hu, D., 2021. HIF-1 α enhances vascular endothelial cell permeability through degradation and translocation of vascular endothelial cadherin and claudin-5 in rats with burn injury. *J. Burn Care Res.* 42 (2), 258–268. <https://doi.org/10.1093/jbcr/iraa139>.
- John, S., 2020. Lessons learned from kidney dysfunction Preventing organ failure. *Med. Klin. Intensivmed. Notfallmed.* 7. <https://doi.org/10.1007/s00063-020-00659-2>.
- Kwon, M.Y., Kim, S.M., Park, J., Lee, J., Cho, H., Lee, H., Jeon, C., Park, J.H., Han, I.O., 2019. A caffeic acid-ferulic acid hybrid compound attenuates lipopolysaccharide-mediated inflammation in BV2 and RAW264.7 cells. *Biochem. Biophys. Res. Commun.* 515 (4), 565–571. <https://doi.org/10.1016/j.bbrc.2019.06.005>.
- Lavrentieva, A., Depetris, N., Moiemien, N., Joannidis, M., Palmieri, T.L., 2022. Renal replacement therapy for acute kidney injury in burn patients, an international survey and a qualitative review of current controversies. *Burns* 48 (5), 1079–1091. <https://doi.org/10.1016/j.burns.2021.08.013>.
- Li, H., Yan, Z., Xiong, Q., Chen, X., Lin, Y., Xu, Y., Bai, L., Jiang, W., Zheng, D., Xing, C., 2019. Renoprotective effect and mechanism of polysaccharide from *Polyporus umbellatus* sclerotia on renal fibrosis. *Carbohydr. Polym.* 212, 1–10. <https://doi.org/10.1016/j.carbpol.2019.02.026>.
- Li, H.D., Meng, X.M., Huang, C., Zhang, L., Lv, X.W., Li, J., 2019. Application of herbal traditional Chinese medicine in the treatment of acute kidney injury. *Front. Pharmacol.* 10, 12. <https://doi.org/10.3389/fphar.2019.00376>.
- Li, H.S., Yao, Z.H., Tan, J.L., Zhou, J.Y., Li, Y., Wu, J., Luo, G.X., 2017. Epidemiology and outcome analysis of 6325 burn patients: a five-year retrospective study in a major burn center in Southwest China. *Sci. Rep.* 7, 9. <https://doi.org/10.1038/srep46066>.
- Li, S., Zhang, B., 2013. Traditional Chinese medicine network pharmacology: theory, methodology and application. *Chin. J. Nat. Med.* 11 (2), 110–120. [https://doi.org/10.1016/s1875-5364\(13\)60037-0](https://doi.org/10.1016/s1875-5364(13)60037-0).
- Li, Y.Z., Ren, S., Yan, X.T., Li, H.P., Li, W., Zheng, B., Wang, Z., Liu, Y.Y., 2018. Improvement of Cisplatin-induced renal dysfunction by *Schisandra chinensis* stems via anti-inflammation and anti-apoptosis effects. *J. Ethnopharmacol.* 217, 228–237. <https://doi.org/10.1016/j.jep.2018.01.033>.
- Liang, D., Yong, T., Diao, X., Chen, S., Chen, D., Xiao, C., Zuo, D., Xie, Y., Zhou, X., Hu, H., 2021. Hypouricaemic and nephroprotective effects of *Poria cocos* in hyperuricemic mice by up-regulating ATP-binding cassette super-family G member 2. *Pharm. Biol.* 59 (1), 275–286. <https://doi.org/10.1080/13880209.2021.1885450>.
- Liu, C., Zhao, S., Zhu, C., Gao, Q., Bai, J., Si, J., Chen, Y., 2020. Ergosterol ameliorates renal inflammatory responses in mice model of diabetic nephropathy. *Biomed. Pharmacother.* 128, 110252. <https://doi.org/10.1016/j.biopha.2020.110252>.
- Liu, J., Yoshida, Y., Yamashita, U., 2008. DNA-binding activity of NF- κ B and phosphorylation of p65 are induced by N-acetylcysteine through phosphatidylinositol (PI) 3-kinase. *Mol. Immunol.* 45 (15), 3984–3989. <https://doi.org/10.1016/j.molimm.2008.06.012>.
- Liu, Q., Lin, X., Li, H., Yuan, J., Peng, Y., Dong, L., Dai, S., 2016. Paeoniflorin ameliorates renal function in cyclophosphamide-induced mice via AMPK suppressed inflammation and apoptosis. *Biomed. Pharmacother.* 84, 1899–1905. <https://doi.org/10.1016/j.biopha.2016.10.097>.
- Liu, Z.G., 2005. Molecular mechanism of TNF signaling and beyond. *Cell Res.* 15 (1), 24–27. <https://doi.org/10.1038/sj.cr.7290259>.
- Luan, Z.L., Ming, W.H., Sun, X.W., Zhang, C., Zhou, Y., Zheng, F., Yang, Y.L., Guan, Y.F., Zhang, X.Y., 2021. A naturally occurring FXR agonist, alisol B 23-acetate, protects against renal ischemia-reperfusion injury. *Am. J. Physiol. Ren. Physiol.* 321 (5), F617–f628. <https://doi.org/10.1152/ajprenal.00193.2021>.
- Mayhall, C.G., 2003. The epidemiology of burn wound infections: then and now. *Clin. Infect. Dis.* 37 (4), 543–550. <https://doi.org/10.1086/376993>.
- Niculae, A., Peride, I., Tiglis, M., Sharkov, E., Neagu, T.P., Lascar, I., Checherita, I.A., 2022. Burn-induced acute kidney injury-two-lane road: from molecular to clinical aspects. *Int. J. Mol. Sci.* 23 (15), 16. <https://doi.org/10.3390/ijms23158712>.
- Nogales, C., Mamdouh, Z.M., List, M., Kiel, C., Casas, A.I., Schmidt, H., 2022. Network pharmacology: curing causal mechanisms instead of treating symptoms. *Trends Pharmacol. Sci.* 43 (2), 136–150. <https://doi.org/10.1016/j.tips.2021.11.004>.
- Noor, F., Tahir, U. Qamar, M., Ashfaq, U.A., Albutti, A., Alwashmi, A.S.S., Aljasir, M.A., 2022. Network pharmacology approach for medicinal plants: review and assessment. *Pharmaceuticals* 15 (5). <https://doi.org/10.3390/ph15050572>.
- Oeckinghaus, A., Hayden, M.S., Ghosh, S., 2011. Crosstalk in NF- κ B signaling pathways. *Nat. Immunol.* 12 (8), 695–708. <https://doi.org/10.1038/ni.2065>.
- Palmieri, T., Lavrentieva, A., Greenhalgh, D.G., 2010. Acute kidney injury in critically ill burn patients. Risk factors, progression and impact on mortality. *Burns* 36 (2), 205–211. <https://doi.org/10.1016/j.burns.2009.08.012>.
- Prowle, J.R., Kirwan, C.J., Bellomo, R., 2014. Fluid management for the prevention and attenuation of acute kidney injury. *Nat. Rev. Nephrol.* 10 (1), 37–47. <https://doi.org/10.1038/nrneph.2013.232>.
- Rabb, H., Griffin, M.D., McKay, D.B., Swaminathan, S., Pickkers, P., Rosner, M.H., Kellum, J.A., Ronco, C., Acute Dialysis Quality, I., 2016. Inflammation in AKI: current understanding, key questions, and knowledge gaps. *J. Am. Soc. Nephrol.* 27 (2), 371–379. <https://doi.org/10.1681/asn.2015030261>.
- Ramesh, G., Reeves, W.B., 2002. TNF- α mediates chemokine and cytokine expression and renal injury in cisplatin nephrotoxicity. *J. Clin. Invest.* 110 (6), 835–842. <https://doi.org/10.1172/jci15606>.
- Ran, J., Ma, C., Xu, K., Xu, L., He, Y., Moqbel, S.A.A., Hu, P., Jiang, L., Chen, W., Bao, J., Xiong, Y., Wu, L., 2018. Schisandrin B ameliorated chondrocytes inflammation and osteoarthritis via suppression of NF- κ B and MAPK signal pathways. *Drug Des. Dev. Ther.* 12, 1195–1204. <https://doi.org/10.2147/dddt.S162014>.
- Sarkar, F.H., Li, Y.W., Wang, Z.W., Kong, D.J., 2008. NF- κ B signaling pathway and its therapeutic implications in human diseases. *Int. Rev. Immunol.* 27 (5), 293–319. <https://doi.org/10.1080/08830180802276179>.
- Seo, E.J., Fischer, N., Efferth, T., 2018. Phytochemicals as inhibitors of NF- κ B for treatment of Alzheimer's disease. *Pharmacol. Res.* 129, 262–273. <https://doi.org/10.1016/j.phrs.2017.11.030>.
- Sharma, B., Yadav, D.K., 2022. Metabolomics and network pharmacology in the exploration of the multi-targeted therapeutic approach of traditional medicinal plants. *Plants* 11 (23). <https://doi.org/10.3390/plants11233243>.
- Shaterzadeh-Yazdi, H., Noorbakhsh, M.F., Samarghandian, S., Farkhondeh, T., 2018. An overview on renoprotective effects of thymoquinone. *Kidney Dis.* 4 (2), 74–82. <https://doi.org/10.1159/000486829>.
- Shu, Y., Yang, Y.Y., Zhao, Y.L., Ma, L., Fu, P., Wei, T.T., Zhang, L., 2019. Melittin inducing the apoptosis of renal tubule epithelial cells through upregulation of bax/bcl-2 expression and activation of TNF- α signaling pathway. *BioMed Res. Int.* 13, 2019. <https://doi.org/10.1155/2019/9450368>.
- Tang, J.L., Xin, M., Zhang, L.C., 2022. Protective effect of *Astragalus membranaceus* and Astragaloside IV in sepsis-induced acute kidney injury. *Aging (Albany NY)* 14 (14), 5855–5877. <https://doi.org/10.18632/aging.204189>.
- Villa, G., Husain-Syed, F., Saitta, T., Degl'Innocenti, D., Barbani, F., Resta, M., Castellani, G., Romagnoli, S., 2021. Hemodynamic instability during acute kidney injury and acute renal replacement therapy: pathophysiology and clinical implications. *Blood Purif.* 50 (6), 729–739. <https://doi.org/10.1159/000513942>.
- Walker, M.R., Leung, P., Eltahla, A.A., Underwood, A., Abayasingam, A., Brasher, N.A., Li, H., Wu, B.R., Maher, L., Luciani, F., Lloyd, A.R., Bull, R.A., 2019. Clearance of hepatitis C virus is associated with early and potent but narrowly-directed, Envelope-specific antibodies. *Sci. Rep.* 9, 14. <https://doi.org/10.1038/s41598-019-49454-w>.
- Wang, Q.H., Kuang, H.X., Su, Y., Sun, Y.P., Feng, J., Guo, R., Chan, K., 2013. Naturally derived anti-inflammatory compounds from Chinese medicinal plants. *J. Ethnopharmacol.* 146 (1), 9–39. <https://doi.org/10.1016/j.jep.2012.12.013>.

- Witkowski, W., Kawecki, M., Surowiecka-Pastewka, A., Klimm, W., Szamotulska, K., Niemczyk, S., 2016. Early and late acute kidney injury in severely burned patients. *Med. Sci. Mon. Int. Med. J. Exp. Clin. Res.* 22, 3755–3763. <https://doi.org/10.12659/msm.895875>.
- Wu, X.Y., Guo, R.Q., Chen, P.L., Wang, Q., Cunningham, P.N., 2009. TNF induces caspase-dependent inflammation in renal endothelial cells through a Rho- and myosin light chain kinase-dependent mechanism. *Am. J. Physiol. Ren. Physiol.* 297 (2), F316–F326. <https://doi.org/10.1152/ajprenal.00089.2009>.
- Xu, Y., Hu, X., Cai, J., Li, Y., Zou, Y., Wang, Y., Xie, C., Xu, S., Wang, Y., Zheng, Y., Mahamat, D.A., Xu, Y., Wang, X., Li, X., Liu, A., Chen, D., Zhu, L., Guo, J., 2023. Atractylenolide-III alleviates osteoarthritis and chondrocyte senescence by targeting NF- κ B signaling. *Phytother. Res.* <https://doi.org/10.1002/ptr.7929>.
- Yan, J.J., Ryu, J.H., Piao, H., Hwang, J.H., Han, D., Lee, S.K., Jang, J.Y., Lee, J., Koo, T. Y., Yang, J., 2020. Granulocyte colony-stimulating factor attenuates renal ischemia-reperfusion injury by inducing myeloid-derived suppressor cells. *J. Am. Soc. Nephrol.* 31 (4), 731–746. <https://doi.org/10.1681/asn.2019060601>.
- Yang, L., Yu, H., Hou, A., Man, W., Wang, S., Zhang, J., Wang, X., Zheng, S., Jiang, H., Kuang, H., 2021. A review of the ethnopharmacology, phytochemistry, pharmacology, application, quality control, processing, toxicology, and pharmacokinetics of the dried rhizome of *Atractylodes macrocephala*. *Front. Pharmacol.* 12, 727154. <https://doi.org/10.3389/fphar.2021.727154>.
- Yi, L., Lu, Y., Yu, S., Cheng, Q., Yi, L., 2022. Formononetin inhibits inflammation and promotes gastric mucosal angiogenesis in gastric ulcer rats through regulating NF- κ B signaling pathway. *J. Recept. Signal Transduct. Res.* 42 (1), 16–22. <https://doi.org/10.1080/10799893.2020.1837873>.
- Yokozawa, T., Kim, H.Y., Yamabe, N., 2004. Amelioration of diabetic nephropathy by dried rehmanniae radix (di huang) extract. *Am. J. Chin. Med.* 32 (6), 829–839. <https://doi.org/10.1142/s0192415x04002442>.
- Yoshimura, A., Mori, H., Ohishi, M., Aki, D., Hanada, T., 2003. Negative regulation of cytokine signaling influences inflammation. *Curr. Opin. Immunol.* 15 (6), 704–708. <https://doi.org/10.1016/j.coi.2003.09.004>.
- Yuan, C.Y., Wang, Q.C., Chen, X.L., Wang, Q., Sun, C.S., Sun, Y.X., Wang, C.H., Su, M.X., Wang, H.Y., Wu, X.S., 2019. Hypertonic saline resuscitation protects against kidney injury induced by severe burns in rats. *Burns* 45 (3), 641–648. <https://doi.org/10.1016/j.burns.2018.09.017>.
- Zeng, X., Chen, X., Qin, H., Han, Y., Chen, X., Han, Z., Zhao, W., 2020. Preventive effects of a natural anti-inflammatory agent Salvianolic acid A on acute kidney injury in mice. *Food Chem. Toxicol.* 135, 110901. <https://doi.org/10.1016/j.fct.2019.110901>.
- Zhang, R., Zhu, X., Bai, H., Ning, K., 2019. Network pharmacology databases for traditional Chinese medicine: review and assessment. *Front. Pharmacol.* 10, 123. <https://doi.org/10.3389/fphar.2019.00123>.
- Zhao, H., Xu, J., Wang, R., Tang, W., Kong, L., Wang, W., Wang, L., Zhang, Y., Ma, W., 2021. Plantaginis Semen polysaccharides ameliorate renal damage through regulating NLRP3 inflammasome in gouty nephropathy rats. *Food Funct.* 12 (6), 2543–2553. <https://doi.org/10.1039/d0fo03143g>.
- Zhao, J., Duan, Q.X., Dong, C.H., Cui, J., 2022. CUL4A ATTENUATES LPS-INDUCED ACUTE KIDNEY INJURY VIA BLOCKING NF- κ B SIGNALING PATHWAY IN SEPSIS. *J. Med. Biochem.* 41 (1), 62–70. <https://doi.org/10.5937/jomb0-33096>.
- Zhao, L., Zhang, H., Li, N., Chen, J., Xu, H., Wang, Y., Liang, Q., 2023. Network pharmacology, a promising approach to reveal the pharmacology mechanism of Chinese medicine formula. *J. Ethnopharmacol.* 309, 116306. <https://doi.org/10.1016/j.jep.2023.116306>.